Python Practice Questions with Answers

1. Create variables for: Name, Age, Phone number, Address. Display all.

```
name = "Binty"
age = 25
phone = "9876543210"
address = "Kerala"

print("Name:", name)
print("Age:", age)
print("Phone:", phone)
print("Address:", address)
```

2. Store two numbers and find: Addition, Subtraction, Multiplication, Division.

```
a = 10
b = 5

print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
```

3. Swap two variables.

```
x = 20
y = 30

print("Before Swapping")
print("x =", x)
print("y =", y)

temp = x
x = y
y = temp

print("After Swapping")
print("x =", x)
print("y =", y)
```

4. Find area of Circle.

```
radius = 7
circle_area = 3.14 * radius * radius

print("Area of Circle =", circle_area)
```

5. Find area of Rectangle.

```
length = 10
breadth = 5
rectangle_area = length * breadth

print("Area of Rectangle =", rectangle_area)
```

6. Find area of Triangle.

```
base = 8
height = 6
triangle_area = 0.5 * base * height

print("Area of Triangle =", triangle_area)
```

7. Calculate Simple Interest.

```
principal = 1000
rate = 5
time = 2

simple_interest = (principal * rate * time) / 100

print("Simple Interest =", simple_interest)
```

8. Calculate Compound Interest.

```
principal = 1000
rate = 5
time = 2

amount = principal * (1 + rate / 100) ** time
compound_interest = amount - principal

print("Compound Interest =", compound_interest)
```

9. Convert Celsius to Fahrenheit.

```
celsius = 37
fahrenheit = (celsius * 9/5) + 32

print("Fahrenheit =", fahrenheit)
```

10. Convert Kilometers to Meters.

```
km = 5
meters = km * 1000

print("Meters =", meters)
```

11. Convert Minutes to Hours.

```
minutes = 120
hours = minutes / 60

print("Hours =", hours)
```

12. Take user input: Name, Age, College and display output.

```
user_name = input("Enter your name: ")
user_age = input("Enter your age: ")
college = input("Enter your college name: ")

print("--- User Details ---")
print("Name:", user_name)
print("Age:", user_age)
print("College:", college)
```

Python Practice Questions and Answers Operators and Control Statements

1. Write a program to perform all arithmetic operations on two numbers.

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Addition:", num1 + num2)
print("Subtraction:", num1 - num2)
print("Multiplication:", num1 * num2)
print("Division:", num1 / num2)
print("Modulus:", num1 % num2)
```

2. Check whether a number is odd or even using operators.

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

3. Calculate simple interest using principal, rate, and time.

```
p = float(input("Enter principal amount: "))
r = float(input("Enter rate: "))
t = float(input("Enter time: "))

si = (p * r * t) / 100
```

```
print("Simple Interest =", si)
```

4. Check whether a person is eligible to vote based on age.

```
age = int(input("Enter age: "))

if age >= 18:
    print("Eligible to vote")
else:
    print("Not eligible")
```

5. Increase an employee's salary by 10% using assignment operators.

```
salary = float(input("Enter salary: "))

salary += salary * 0.10

print("Updated Salary =", salary)
```

6. Check if a student passed all subjects using logical operators.

```
m1 = int(input("Enter mark1: "))
m2 = int(input("Enter mark2: "))
m3 = int(input("Enter mark3: "))

if m1 >= 40 and m2 >= 40 and m3 >= 40:
    print("Passed all subjects")
else:
    print("Failed")
```

7. Find the largest among three numbers using comparison operators.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))

if a > b and a > c:
    print("Largest =", a)
elif b > c:
    print("Largest =", b)
else:
    print("Largest =", c)
```

8. Create an ATM withdrawal system and update balance after withdrawal.

```
balance = 10000

withdraw = float(input("Enter withdrawal amount: "))
```

```
if withdraw <= balance:
    balance -= withdraw
    print("Withdrawal successful")
    print("Remaining Balance =", balance)
else:
    print("Insufficient balance")
```

9. Check whether a username exists in a list using membership operators.

```
users = ["binty", "rahul", "anu"]

name = input("Enter username: ")

if name in users:
    print("Username exists")
else:
    print("Username not found")
```

10. Calculate BMI and determine the health category.

```
weight = float(input("Enter weight in kg: "))
height = float(input("Enter height in meters: "))

bmi = weight / (height ** 2)

print("BMI =", bmi)

if bmi < 18.5:
    print("Underweight")
elif bmi < 25:
    print("Normal")
elif bmi < 30:
    print("Overweight")
else:
    print("Obese")
```

11. Create an electricity bill calculator based on units consumed.

```
units = int(input("Enter units consumed: "))

if units <= 100:
    bill = units * 5
elif units <= 200:
    bill = units * 7
else:
    bill = units * 10

print("Electricity Bill =", bill)
```

12. Check loan eligibility using age, salary, and employment status.

```
age = int(input("Enter age: "))
salary = float(input("Enter salary: "))
job = input("Are you employed? (yes/no): ")

if age >= 21 and salary >= 25000 and job == "yes":
    print("Eligible for loan")
else:
    print("Not eligible")
```

13. Compare two strings and determine whether they are equal.

```
str1 = input("Enter first string: ")
str2 = input("Enter second string: ")

if str1 == str2:
    print("Strings are equal")
else:
    print("Strings are not equal")
```

14. Check whether a character exists in a given string.

```
text = input("Enter a string: ")
char = input("Enter a character: ")

if char in text:
    print("Character exists")
else:
    print("Character not found")
```

15. Compare two lists using == and is operators.

```
list1 = [1, 2, 3]
list2 = [1, 2, 3]

print(list1 == list2)
print(list1 is list2)
```

16. Calculate shopping discount based on purchase amount.

```
amount = float(input("Enter purchase amount: "))

if amount >= 5000:
    discount = amount * 0.20
elif amount >= 2000:
    discount = amount * 0.10
else:
    discount = 0
```

```
final_amount = amount - discount

print("Discount =", discount)
print("Final Amount =", final_amount)
```

17. Perform bitwise operations (&, |, ^, <<, >>) on two numbers.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("AND =", a & b)
print("OR =", a | b)
print("XOR =", a ^ b)
print("Left Shift =", a << 1)
print("Right Shift =", a >> 1)
```

18. Update mobile balance after recharge using assignment operators.

```
balance = 100

recharge = int(input("Enter recharge amount: "))

balance += recharge

print("Updated Balance =", balance)
```

19. Check employee bonus eligibility using experience and salary conditions.

```
experience = int(input("Enter years of experience: "))
salary = float(input("Enter salary: "))

if experience >= 5 and salary < 50000:
    print("Eligible for bonus")
else:
    print("Not eligible")
```

20. Build a mini calculator to perform arithmetic operations based on user input.

```
num1 = float(input("Enter first number: "))
operator = input("Enter operator (+, -, *, /): ")
num2 = float(input("Enter second number: "))

if operator == "+":
    print("Result =", num1 + num2)
elif operator == "-":
    print("Result =", num1 - num2)
elif operator == "*":
    print("Result =", num1 * num2)
```

```
elif operator == "/":
    print("Result =", num1 / num2)
else:
    print("Invalid operator")
```

Conditional Statements – 10 Questions and Full Answers

1. Write a program to check whether a year is a leap year or not using conditional statements.

```
year = int(input("Enter year: "))

if year % 400 == 0:
    print("Leap Year")
elif year % 100 == 0:
    print("Not Leap Year")
elif year % 4 == 0:
    print("Leap Year")
else:
    print("Not Leap Year")
```

2. Write a program to find the largest among 4 numbers entered by the user.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
d = int(input("Enter fourth number: "))

if a >= b and a >= c and a >= d:
    print("Largest =", a)
elif b >= c and b >= d:
    print("Largest =", b)
elif c >= d:
    print("Largest =", c)
else:
    print("Largest =", d)
```

3. Write a program to check whether a character is uppercase, lowercase, digit or special character.

```
ch = input("Enter a character: ")

if ch >= 'A' and ch <= 'Z':
    print("Uppercase")
elif ch >= 'a' and ch <= 'z':
    print("Lowercase")
elif ch >= '0' and ch <= '9':
    print("Digit")
```

```
else:
    print("Special Character")
```

4. Write a program to calculate electricity bill.

```
units = int(input("Enter units: "))

if units <= 100:
    bill = units * 5
elif units <= 200:
    bill = (100 * 5) + ((units - 100) * 7)
else:
    bill = (100 * 5) + (100 * 7) + ((units - 200) * 10)

print("Bill =", bill)
```

5. Write a program to check whether three sides can form a triangle and find the type.

```
a = int(input("Enter side1: "))
b = int(input("Enter side2: "))
c = int(input("Enter side3: "))

if a + b > c and a + c > b and b + c > a:
    if a == b and b == c:
        print("Equilateral Triangle")
    elif a == b or b == c or a == c:
        print("Isosceles Triangle")
    else:
        print("Scalene Triangle")
else:
    print("Not a Triangle")
```

6. Write a program to input marks of 5 subjects and print grade.

```
m1 = int(input("Enter mark1: "))
m2 = int(input("Enter mark2: "))
m3 = int(input("Enter mark3: "))
m4 = int(input("Enter mark4: "))
m5 = int(input("Enter mark5: "))

total = m1 + m2 + m3 + m4 + m5
avg = total / 5

if m1 < 50 or m2 < 50 or m3 < 50 or m4 < 50 or m5 < 50:
    print("Fail")
elif avg > 90:
    print("Grade A")
elif avg >= 75:
    print("Grade B")
elif avg >= 50:
```

```
    print("Grade C")
else:
    print("Fail")
```

7. Write a program to check whether a number is positive even, positive odd, negative even, negative odd or zero.

```
num = int(input("Enter number: "))

if num == 0:
    print("Zero")
elif num > 0:
    if num % 2 == 0:
        print("Positive Even")
    else:
        print("Positive Odd")
else:
    if num % 2 == 0:
        print("Negative Even")
    else:
        print("Negative Odd")
```

8. Write a program to create a simple login system.

```
username = input("Enter username: ")
password = input("Enter password: ")

if username == "admin":
    if password == "1234":
        print("Login Successful")
    else:
        print("Incorrect Password")
else:
    print("User Not Found")
```

9. Write a program to check eligibility for voting, driving license and senior citizen discount.

```
age = int(input("Enter age: "))

if age >= 18:
    print("Eligible for voting")
else:
    print("Not eligible for voting")

if age >= 18:
    print("Eligible for driving license")
else:
    print("Not eligible for driving license")

if age >= 60:
```

```
    print("Eligible for senior citizen discount")
else:
    print("Not eligible for senior citizen discount")
```

10. Write a program for a mini ATM system.

```
pin = int(input("Enter PIN: "))
balance = 5000

if pin == 1234:
    print("PIN Correct")
    amount = int(input("Enter withdrawal amount: "))

    if amount <= balance:
        balance = balance - amount
        print("Withdrawal successful")
        print("Balance =", balance)
    else:
        print("Insufficient Balance")
else:
    print("Wrong PIN")
```